

# Informative Power-System Probing via Sequential Bayesian Experimental Design

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**Abstract**—Power-grid stability is vital to industrial operations and daily life. Understanding uncertain inertia and frequency-response parameters supports stability assessment and control. Active probing obtains information about these parameters by injecting small signals and measuring the response. However, predetermined sequences cannot exploit new measurements, while myopic design considers only next-step information gain and requires online optimization. We apply sequential Bayesian optimal experimental design to learn adaptive policies that select probe durations from observed responses and optimize information gain over the complete sequence. On reduced IEEE9 and IEEE14 models, deep adaptive design (DAD), its online-refined variant Step-DAD, and reinforcement learning (RL-sBOED) achieve higher mean estimated information gain than Myopic, an optimized fixed sequence, and random probing at equal probe budgets. Offline-trained DAD and RL-sBOED also substantially reduce online decision time relative to Myopic. Step-DAD provides modest additional gains on IEEE9, with substantial online cost, but no consistent improvement over DAD on IEEE14.

**Index Terms**—Bayesian experimental design, power-system identification, adaptive probing, expected information gain, rate of change of frequency.

## I. INTRODUCTION

Electrical power systems connect generators and consumers through transmission networks. A disturbance to the balance between generation and demand changes system frequency; understanding this response is essential to maintaining the reliable supply required by industry, public services, and households. Inertia resists rapid frequency changes, while frequency-response mechanisms adjust power following frequency deviations [1]. In the reduced models considered here, these effects are represented by inertia  $M$  and effective frequency-response coefficients  $K$ . These parameters are latent: their values must be inferred from measured behavior. Reducing their uncertainty supports predictions of disturbance responses and informed control decisions.

Active probing is an established approach to obtaining such information. A small, known power signal is injected into the system, and the resulting frequency response is measured [1]. The response provides evidence about plausible parameter values, but the information obtained depends on the signal design. For example, pulse duration changes the transient response and the evidence available for identification. This raises an experimental-design question: how should a limited

number of probes be selected to learn as much as possible about the unknown parameters?

Bayesian optimal experimental design (BOED) provides a framework for this question [2]. A prior distribution describes parameter uncertainty before an experiment. Bayes' rule combines this prior with the measurement model and observed response to form a posterior distribution. Expected information gain (EIG) quantifies the average reduction in uncertainty anticipated from an experiment, accounting for unknown parameters and noisy outcomes. It therefore provides a criterion for choosing a probe before its response is available.

For a sequence of experiments, fixed and greedy designs have distinct limitations. A fixed sequence cannot use incoming measurements to revise later probes. A greedy, or myopic, design uses observation feedback but maximizes only the next step's EIG, without accounting for subsequent learning opportunities [2]. Repeated online optimization can also make design selection expensive. The challenge is to combine observation feedback with planning over the complete sequence while limiting online computation.

Policy-based sequential BOED addresses this challenge. Early measurements inform later choices, and training over complete sequences accounts for the value of future experiments. Deep adaptive design (DAD) [3] and reinforcement-learning-based sequential BOED (RL-sBOED) [4] learn a mapping from past designs and observations to the next design. Offline training amortizes optimization by reusing this mapping during experiments. Step-DAD [5] additionally refines a pretrained policy as observations arrive, allowing the value of further adaptation to be weighed against its online cost.

We apply these existing methods to probe-duration selection on reduced IEEE9 and IEEE14 models. These benchmark networks have nine and fourteen buses, respectively: electrical connection points for lines, generation, or demand [6]. The reduced models retain coupled machine dynamics, and physical state carries between probes. Our contribution is the sequential grid-probing formulation and a matched comparison of DAD, RL-sBOED, Step-DAD, Myopic, an optimized fixed sequence, and random probing. We evaluate the information obtained at equal probe budgets and separately measure offline training and online decision costs.

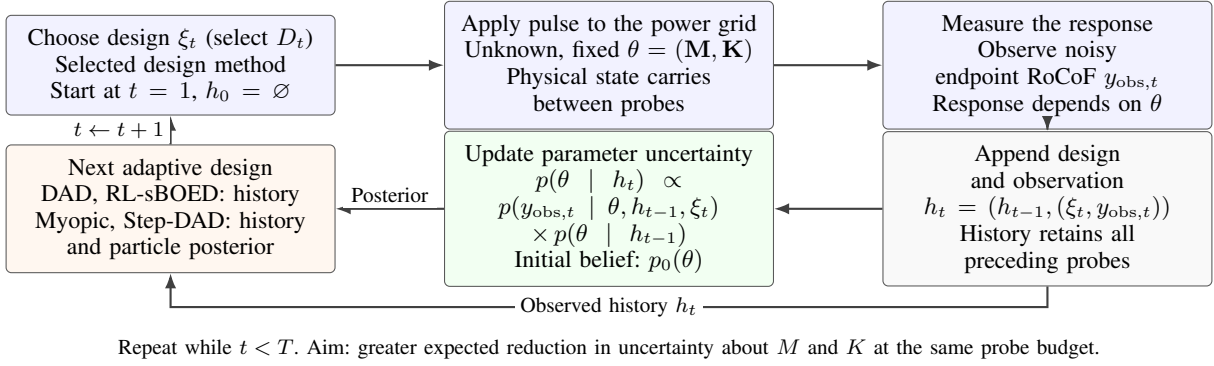


Fig. 1. Sequential probing with physical state carried between experiments. Observations update the design history and parameter uncertainty. DAD/RL-sBOED use history directly; Myopic/Step-DAD additionally maintain a particle posterior. Fixed and Random bypass observation feedback. The loop ends after  $T$  measurements.

## II. PROBLEM FORMULATION

We seek to reduce uncertainty in  $M$  and  $K$  through informative probes within a budget of  $T$  design steps. Here  $t = 1, \dots, T$  indexes design steps,  $d$  denotes elapsed time within each probe window, and  $D_t$  is the selected pulse duration.

### A. Unknown Parameters and Power-System Model

For a reduced network with  $m$  machines, the unknown parameter vector is  $\theta = (M_1, \dots, M_m, K_1, \dots, K_m)$ , where  $M_i$  and  $K_i$  are inertia and effective frequency-response coefficients. The prior  $p_0(\theta)$  has independent components  $M_i \sim \mathcal{U}(0.7\bar{M}_i, 1.3\bar{M}_i)$  and  $K_i \sim \mathcal{U}(0.1\bar{K}_i, \bar{K}_i)$ . One draw of  $\theta$  remains fixed throughout all  $T$  probes.

Let  $\delta_i$  be rotor angle,  $\omega_i$  angular-speed deviation, and  $u_i$  the applied probing power. The reduced swing model is

$$\begin{aligned} \dot{\delta}_i &= \omega_i, \\ M_i \dot{\omega}_i &= u_i - \left( D_i + \frac{K_i}{2\pi} \right) \omega_i \\ &\quad - \sum_{j=1}^m B_{ij} \sin(\delta_i - \delta_j). \end{aligned} \quad (1)$$

Dots denote derivatives with respect to physical time  $d$ . The damping  $D_i$  and network couplings  $B_{ij}$  are known; frequency deviation is  $\Delta f_i = \omega_i/(2\pi)$  in Hz. The units of  $M_i$ ,  $K_i$  and  $D_i$  are  $\text{pu s}^2/\text{rad}$ ,  $\text{pu/Hz}$  and  $\text{pu s/rad}$ , respectively, on a 100-MVA base at nominal frequency 60 Hz. All angles and speed deviations start at zero. Net mechanical injections are zero in this deviation model, making the initial unforced state an equilibrium.

IEEE9 retains buses 1, 2 and 3; IEEE14 retains buses 1, 2, 3, 6 and 8. Test-case reactances [6] define a lossless Laplacian  $\mathbf{L}$ . Eliminating algebraic buses  $l$  while retaining machines  $g$  gives  $\mathbf{L}_{\text{red}} = \mathbf{L}_{gg} - \mathbf{L}_{gl}\mathbf{L}_{ll}^{-1}\mathbf{L}_{lg}$ , with  $B_{ij} = -(\mathbf{L}_{\text{red}})_{ij}$  for  $i \neq j$  and  $B_{ii} = 0$ . Resistance, voltage dynamics and transformer tap effects are omitted. Table I summarizes the model parameterization. Nominal inertia is computed as  $\bar{M}_i = 2H_i/(2\pi f_0)$ , with  $f_0 = 60$  Hz. For IEEE9,  $H = (23.64, 6.40, 3.01)$  s

TABLE I  
PRIOR SCALES AND DAMPING COEFFICIENTS

| Network | Bus     | $\bar{M}_i$ | $\bar{K}_i$ | $D_i$     |
|---------|---------|-------------|-------------|-----------|
| IEEE9   | 1       | 0.1254141   | 0.2937500   | 0.0125414 |
|         | 2       | 0.0339531   | 0.1312500   | 0.0033953 |
|         | 3       | 0.0159685   | 0.0750000   | 0.0015969 |
| IEEE14  | 1       | 0.0212207   | 0.0784314   | 0.0021221 |
|         | 2       | 0.0344836   | 0.1274510   | 0.0034484 |
|         | 3, 6, 8 | 0.0265257   | 0.0980392   | 0.0026526 |

matches published machine data [7, Table 2]. For IEEE14,  $H = (4, 6.5, 5, 5, 5)$  s is prescribed at buses (1, 2, 3, 6, 8) as a benchmark setting, without a verified literature source for these values. For both networks,  $\bar{K}_i = K_{\text{tot}} w_i$ , with  $K_{\text{tot}} = 0.5$  pu/Hz. The allocation weights are  $w = (0.5875, 0.2625, 0.15)$  for IEEE9 and  $w = (8, 13, 10, 10, 10)/51$  for IEEE14. Fixed damping satisfies  $D_i = (0.1 \text{ s}^{-1}) \bar{M}_i$ . The response scales, damping and prior ranges are shared study assumptions, not validated operating-grid parameters. Table entries are rounded; configurations retain full precision. Training and evaluation draw independently from the same prior.

### B. Probe Design and Observation Model

The probe design is  $\xi_t = (A_t, B_t, D_t)$ : amplitude, injection bus and pulse duration. This study fixes  $A_t = 0.05$  pu and  $B_t = 1$ ; only  $D_t$  is optimized. The raised-cosine input for  $0 \leq d \leq D_t$  is

$$u_{B_t,t}(d) = \frac{A_t}{2} [1 - \cos(2\pi d/D_t)]. \quad (2)$$

The input is zero for  $D_t < d \leq D_{\text{obs}}$  and at all other buses. The observation window is  $D_{\text{obs}} = 3.5$  s. The full angle and speed state at its end becomes the initial state of the next window, without resetting the system. Consequently, the response depends on the complete design sequence  $\xi_{1:t} = (\xi_1, \dots, \xi_t)$ .

Write  $\Delta f_{1,t}(d; \theta, \xi_{1:t})$  for the resulting bus-1 frequency deviation. The noise-free response is the signed endpoint rate

of change of frequency (RoCoF), denoted by  $y_t$ :

$$y_t(\theta, \xi_{1:t}) = \frac{\Delta f_{1,t}(D_{\text{obs}}) - \Delta f_{1,t}(D_{\text{obs}} - \Delta d_{\text{samp}})}{\Delta d_{\text{samp}}}. \quad (3)$$

The frequency arguments  $\theta, \xi_{1:t}$  are suppressed in the numerator. With sampling interval  $\Delta d_{\text{samp}} = 0.025$  s, the measured response follows

$$y_{\text{obs},t} = y_t(\theta, \xi_{1:t}) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2), \quad (4)$$

where  $\sigma = 0.005$  Hz/s and noise is independent across windows. Each observation becomes available at window end. Decision latency is idealized as zero in simulated time.

### C. Sequential Decisions and Bayesian Updating

Figure 1 summarizes the experiment–inference loop. The data after step  $t$  are the ordered design–observation history  $h_t = ((\xi_k, y_{\text{obs},k}))_{k=1}^t$ , with  $h_0 = \emptyset$ . A policy  $\pi$  selects  $\xi_t$  using only observed history  $h_{t-1}$ , with amplitude and bus fixed as above. It has no access to true parameters or physical state. Feasible durations satisfy

$$\begin{aligned} 0.2 \leq D_t \leq 3, \quad t = 1, \dots, T, \\ D_t - D_{t-1} \geq \Delta D_{\min}, \quad t = 2, \dots, T, \end{aligned} \quad (5)$$

where  $\Delta D_{\min} = 0.01$  s. At step  $t$ , the upper limit is tightened to  $3 - (T - t)\Delta D_{\min}$  to leave room for the remaining probes; the lower limit is 0.2 initially and  $D_{t-1} + \Delta D_{\min}$  thereafter. All methods map a unit action into this interval without sorting or discretizing durations. Let  $\Pi$  denote the policies satisfying these constraints; any policy randomness is independent of  $\theta$ .

The observation model defines a Gaussian likelihood  $p(y_{\text{obs},t} | \theta, h_{t-1}, \xi_t)$  with mean  $y_t(\theta, \xi_{1:t})$  and variance  $\sigma^2$ . Bayes' rule updates parameter uncertainty after each measurement:

$$p(\theta | h_t) = \frac{p(y_{\text{obs},t} | \theta, h_{t-1}, \xi_t) p(\theta | h_{t-1})}{p(y_{\text{obs},t} | h_{t-1}, \xi_t)}. \quad (6)$$

The denominator is the predictive density, obtained by integrating the numerator over  $\theta$ , and  $p(\theta | h_0) = p_0(\theta)$ . Each candidate parameter vector propagates its own physical state under the observed designs to evaluate the likelihood. Conditional on the history, design selection provides no additional knowledge of  $\theta$ ; policy probabilities therefore cancel from the Bayesian update.

### D. Expected Information Gain and Design Objective

To compare possible design policies, we quantify the uncertainty reduction produced by their observations using the Kullback–Leibler divergence of the terminal posterior from the prior. The expected information gain (EIG) of policy  $\pi$  is

$$\mathcal{I}_T(\pi) = \mathbb{E}_{h_T | \pi} [D_{\text{KL}}(p(\theta | h_T) \| p_0(\theta))]. \quad (7)$$

Here  $D_{\text{KL}}(p \| q) = \int p(\theta) \log[p(\theta)/q(\theta)] d\theta$ ; natural logarithms express information in nats. The expectation averages over parameter draws, measurement noise and any policy randomness. The sequential Bayesian design problem is

$$\pi^* \in \arg \max_{\pi \in \Pi} \mathcal{I}_T(\pi). \quad (8)$$

TABLE II  
COMPARISON OF EXPERIMENTAL DESIGN METHODS

| Method       | Adaptive | Myopic | Optimization |
|--------------|----------|--------|--------------|
| Random       | No       | N/A    | None         |
| Fixed [2]    | No       | No     | Offline      |
| Myopic [2]   | Yes      | Yes    | Online       |
| DAD [3]      | Yes      | No     | Offline      |
| RL-sBOED [4] | Yes      | No     | Offline      |
| Step-DAD [5] | Yes      | No     | Both         |

Equation (8) defines the common information objective; the methods below differ in how they approximate or forgo its optimization. All receive the same number of probes. Computation is assessed separately.

Exact EIG requires marginal likelihoods that are costly to evaluate here. We use the sequential prior-contrastive (sPCE) lower bound [3] for training and evaluation. Let  $\theta^{(0)} \sim p_0$  generate the observed history and let  $\theta^{(1)}, \dots, \theta^{(L)}$  be independent prior contrasts. For any candidate  $\theta$ , define the observation likelihood along the realized sequence as

$$\mathcal{L}_t(\theta) = \prod_{k=1}^t p(y_{\text{obs},k} | \theta, h_{k-1}, \xi_k). \quad (9)$$

The sampled information score is then

$$\hat{I}_t = \log \frac{\mathcal{L}_t(\theta^{(0)})}{\frac{1}{L+1} \sum_{\ell=0}^L \mathcal{L}_t(\theta^{(\ell)})}. \quad (10)$$

Likelihoods are evaluated in the log domain. Averaging  $\hat{I}_T$  estimates the sPCE lower bound on  $\mathcal{I}_T(\pi)$ , not exact EIG. Individual scores can be negative but cannot exceed  $\log(L+1)$ . With  $L = 128$ , the ceiling is 4.8598 nats; near-ceiling scores can conceal differences in EIG. We report these estimated lower-bound values as information scores throughout the results. The generating parameters are used in simulated scoring, never as policy inputs.

## III. COMPARED DESIGN METHODS

All six methods use the same observation model and feasible duration interval (5). Table II distinguishes observation feedback from greedy optimization: an adaptive method need not be myopic, and a fixed sequence can be optimized for the full horizon. The algorithms are adaptations of existing methods to the carried-state grid simulator.

### A. Nonadaptive Designs: Random and Fixed

Random draws  $z_t \sim \mathcal{U}(0, 1)$  independently at each step and sets  $D_t = D_t^{\min} + (D_t^{\max} - D_t^{\min})z_t$ . Here  $D_1^{\min} = 0.2$ ,  $D_t^{\min} = D_{t-1} + \Delta D_{\min}$  for  $t > 1$ , and  $D_t^{\max} = 3 - (T - t)\Delta D_{\min}$ . Thus sampling is uniform within the remaining feasible interval, not over complete ordered sequences. It ignores observations and requires no offline training.

Fixed is a static Bayesian design: all durations are optimized jointly before observations are collected [2, Sec. 2.4]. The ordered sequence is optimized using the full-sequence sPCE

objective. It compares short, central, long and spread interior initializations using validation, then optimizes the selected initialization. Each training seed yields a separately fitted sequence. During evaluation, Fixed executes that sequence without adapting to observations.

### B. Online One-Step Design: Myopic

Unlike Fixed, Myopic uses observation feedback. It follows one-step adaptive BOED, maximizing the expected information from the next observation under the current posterior [2, Sec. 2.4]. Our implementation maintains a weighted particle approximation to this posterior. Before each decision after the first, particles propagate their own states under the preceding duration and update likelihood weights using the latest observation. Continuous search then maximizes expected information from the next probe only, using 16 proposals, three search rounds and 32 simulated observations per proposal. Candidate durations obey the same constraints as all other methods. No offline policy is trained.

### C. Offline-Trained Adaptive Policies: DAD and RL-sBOED

DAD and RL-sBOED retain observation feedback while optimizing over the full sequence offline. Both policies receive the stage index and ordered slots containing past durations, scaled observations and availability masks; future slots are masked. They do not receive true parameters or physical states. At evaluation, a forward pass produces the action without online optimization or an explicit particle-posterior update.

DAD [3] uses a deterministic two-layer, 64-unit tanh network, a trainable stage bias and a sigmoid action map. Offline training maximizes sPCE by differentiating through the complete simulated sequence, including observations and carried states. In the joint Fixed/DAD campaign, validation selects between random and interior Fixed-based initialization. The corresponding initialization cost is accounted for in Section V-B.

RL-sBOED [4] uses a REDQ-style ensemble actor-critic adaptation with ten critics and five replay updates per trajectory batch. The actor has two 128-unit ReLU layers. Its per-step information rewards are  $R_t = \hat{I}_t - \hat{I}_{t-1}$ , with  $\hat{I}_0 = 0$  and discount  $\gamma = 1$ , so their sum equals the terminal sPCE score. Training therefore accounts for future design steps without differentiating through the simulator. Entropy regularization uses normalized tanh actions and target entropy  $-1$ ; reported information scores exclude entropy bonuses. Evaluation uses the actor’s deterministic output.

### D. Online Policy Refinement: Step-DAD

Step-DAD [5] tests whether adapting policy weights to the observed history improves on offline training alone. Each episode begins with the matching selected DAD checkpoint and uses it unchanged for the first probe. Before each subsequent decision, it assimilates the latest observation into its 128-particle posterior. It then samples candidate parameters and their carried states from that posterior to simulate the

remaining design steps. These future simulations condition on the actual observed history.

At each refinement, a fresh Adam optimizer performs four gradient updates with batch size 32 and learning rate  $10^{-3}$ . A fixed conditional validation batch compares the unchanged policy and all updated candidates; the best is retained. Retained weights carry to the next design step within the episode, but not to another episode. The total refinement budget is  $4(T - 1)$  updates: 8, 12 and 16 for  $T = 3, 4, 5$ . The planner never receives the evaluator’s true parameters.

For both Myopic and Step-DAD, assimilation is required only before subsequent decisions; neither planner needs a terminal update for the separately evaluated information score.

## IV. EXPERIMENTAL SETTINGS

### A. Simulation and Training Protocol

We compare  $T = 3, 4, 5$  probes on each network, with one observation per probe. Networks share sensor and design constraints but use separate physical parameters and policies.

The swing equations are integrated by RK4 with step  $1/640$  s. Numerical state/duration Jacobians use perturbations of  $10^{-5}$  and include carried-state derivatives. The verification suite compares policy and conditional-refinement gradients with end-to-end finite differences.

Offline training uses 2,000 updates with batch size 32 and three independent training seeds (101, 202 and 303). The DAD and Fixed learning rate is  $10^{-3}$ . Validation uses 128 systems every 100 updates. Training, validation and evaluation use 128 contrastive samples for the sPCE objective. Myopic and Step-DAD use 128 particles for online posterior planning; Step-DAD performs four refinement updates after each observation when design steps remain. Evaluation uses seeds 1001, 1002 and 1003, with 512 simulated experimental sequences per evaluation seed. Validation selects checkpoints without gradient updates on validation cases. Each trained policy is evaluated on 1,536 episodes. Parameter and measurement-noise draws are paired across methods; the online planner uses a separate random stream. Random/Myopic have no training-seed dependence under these streams, so identical evaluations can be shared across training seeds. Fixed remains independently optimized for each training seed.

### B. Evaluation and Computational Measurements

For trained methods, we average evaluation-seed means within each training seed and report mean  $\pm$  sample standard deviation across three trainings. Random/Myopic report mean  $\pm$  sample SD of evaluation-seed means, not episode-level dispersion. Evaluation seed 1001 informed development and is not untouched test data.

Experiments use NVIDIA RTX 5080 and A100 GPUs, with RTX 4090 used for development. IEEE9 offline times combine RTX 5080 and A100 runs; online timings use RTX 5080 except Step-DAD  $T = 5$ , training seed 303, which ran on A100. IEEE14 timings use A100. These recorded workflow costs do not provide a controlled runtime comparison between networks. Configurations, source hashes and

checkpoints preserve provenance. Code and supplementary historical SIR-ODE results are available at [https://github.com/alexschmidt123/tpec2027\\_power\\_grid](https://github.com/alexschmidt123/tpec2027_power_grid).

## V. RESULTS AND DISCUSSION

### A. Information Acquired at Equal Probe Budgets

Tables III and IV report mean sPCE information scores in nats, with the seed-level standard deviations defined above. All three learned methods exceed Random, Fixed, and Myopic in mean score on both networks at every tested horizon.

TABLE III  
IEEE9 INFORMATION SCORE (SPCE, NATS)

| Method   | $T = 3$             | $T = 4$             | $T = 5$             |
|----------|---------------------|---------------------|---------------------|
| Random   | $0.4321 \pm 0.0310$ | $0.4924 \pm 0.0342$ | $0.5392 \pm 0.0471$ |
| Fixed    | $2.0112 \pm 0.0116$ | $2.3192 \pm 0.0004$ | $2.5066 \pm 0.0013$ |
| Myopic   | $1.9736 \pm 0.0438$ | $2.1707 \pm 0.0366$ | $2.2407 \pm 0.0547$ |
| DAD      | $2.1366 \pm 0.0161$ | $2.4095 \pm 0.0080$ | $2.6351 \pm 0.0179$ |
| RL-sBOED | $2.1257 \pm 0.0027$ | $2.3963 \pm 0.0333$ | $2.5307 \pm 0.0114$ |
| Step-DAD | $2.1471 \pm 0.0110$ | $2.4382 \pm 0.0048$ | $2.6621 \pm 0.0145$ |

TABLE IV  
IEEE14 INFORMATION SCORE (SPCE, NATS)

| Method   | $T = 3$             | $T = 4$             | $T = 5$             |
|----------|---------------------|---------------------|---------------------|
| Random   | $0.7541 \pm 0.0047$ | $0.8360 \pm 0.0297$ | $0.8464 \pm 0.0252$ |
| Fixed    | $4.2623 \pm 0.0402$ | $4.4946 \pm 0.0345$ | $4.6624 \pm 0.0083$ |
| Myopic   | $4.2907 \pm 0.0368$ | $4.3990 \pm 0.0177$ | $4.4641 \pm 0.0280$ |
| DAD      | $4.7576 \pm 0.0362$ | $4.8325 \pm 0.0047$ | $4.8443 \pm 0.0026$ |
| RL-sBOED | $4.7445 \pm 0.0022$ | $4.8275 \pm 0.0112$ | $4.8383 \pm 0.0108$ |
| Step-DAD | $4.7527 \pm 0.0275$ | $4.8297 \pm 0.0067$ | $4.8446 \pm 0.0042$ |

On IEEE9, DAD exceeds Fixed by 0.1253, 0.0903, and 0.1286 nats at  $T = 3, 4, 5$ , respectively. Step-DAD improves the DAD mean by 0.0105, 0.0287, and 0.0269 nats. Thus online refinement yields modest additional information in this network, where scores remain well below the estimator ceiling.

On IEEE14, DAD exceeds Myopic by 0.4669 nats and Fixed by 0.4953 nats at  $T = 3$ . Fixed exceeds Myopic at  $T = 4, 5$ , illustrating that observation feedback alone does not guarantee a stronger design when optimization is restricted to next-step gain. The learned methods combine feedback with a full-sequence objective, although this comparison does not isolate the contribution of each mechanism.

Differences among learned policies on IEEE14 are small: Step-DAD changes the DAD mean by  $-0.0049$ ,  $-0.0028$ , and  $+0.0003$  nats at  $T = 3, 4, 5$ . At  $T = 5$ , DAD is only 0.0155 nats below the 4.8598-nat ceiling. The results therefore show no consistent benefit from refinement and cannot establish a reliable ranking of the learned policies' exact EIG.

### B. Measured Computational Cost

Tables V–VIII separate offline training hours from online decision milliseconds per complete sequence. Online timing sums all  $T$  decisions, including posterior updates, Myopic

search, and Step-DAD refinement where applicable. It excludes physical response simulation, observation acquisition, and information scoring. For each evaluation seed, times are averaged over 512 episodes and then over training seeds where applicable; the reported mean and sample SD use the three evaluation-seed means. These are computational costs, not end-to-end experimental durations.

TABLE V  
IEEE9 OFFLINE TRAINING TIME (HOURS)

| Method   | $T = 3$           | $T = 4$           | $T = 5$           |
|----------|-------------------|-------------------|-------------------|
| Random   | 0                 | 0                 | 0                 |
| Fixed    | $0.960 \pm 0.352$ | $1.396 \pm 0.515$ | $1.827 \pm 0.662$ |
| Myopic   | 0                 | 0                 | 0                 |
| DAD      | $0.957 \pm 0.342$ | $1.393 \pm 0.509$ | $1.831 \pm 0.654$ |
| RL-sBOED | $0.120 \pm 0.017$ | $0.152 \pm 0.030$ | $0.180 \pm 0.045$ |
| Step-DAD | $0.957 \pm 0.342$ | $1.393 \pm 0.509$ | $1.831 \pm 0.654$ |

TABLE VI  
IEEE14 OFFLINE TRAINING TIME (HOURS)

| Method   | $T = 3$           | $T = 4$           | $T = 5$           |
|----------|-------------------|-------------------|-------------------|
| Random   | 0                 | 0                 | 0                 |
| Fixed    | $1.877 \pm 0.011$ | $2.750 \pm 0.002$ | $3.691 \pm 0.050$ |
| Myopic   | 0                 | 0                 | 0                 |
| DAD      | $1.894 \pm 0.032$ | $2.725 \pm 0.018$ | $3.626 \pm 0.068$ |
| RL-sBOED | $0.170 \pm 0.008$ | $0.204 \pm 0.001$ | $0.248 \pm 0.001$ |
| Step-DAD | $1.894 \pm 0.032$ | $2.725 \pm 0.018$ | $3.626 \pm 0.068$ |

TABLE VII  
IEEE9 ONLINE DECISION TIME (MS PER SEQUENCE)

| Method   | $T = 3$             | $T = 4$               | $T = 5$           |
|----------|---------------------|-----------------------|-------------------|
| Random   | $0.0715 \pm 0.0001$ | $0.09445 \pm 0.00003$ | $0.121 \pm 0.001$ |
| Fixed    | $0.523 \pm 0.038$   | $0.623 \pm 0.002$     | $0.766 \pm 0.002$ |
| Myopic   | $9308 \pm 4$        | $12577 \pm 2$         | $15892 \pm 2$     |
| DAD      | $0.537 \pm 0.002$   | $0.683 \pm 0.002$     | $0.838 \pm 0.004$ |
| RL-sBOED | $0.492 \pm 0.003$   | $0.626 \pm 0.001$     | $0.764 \pm 0.002$ |
| Step-DAD | $6342 \pm 2$        | $15808 \pm 9$         | $24350 \pm 50$    |

TABLE VIII  
IEEE14 ONLINE DECISION TIME (MS PER SEQUENCE)

| Method   | $T = 3$           | $T = 4$           | $T = 5$           |
|----------|-------------------|-------------------|-------------------|
| Random   | $0.161 \pm 0.004$ | $0.209 \pm 0.002$ | $0.309 \pm 0.049$ |
| Fixed    | $1.16 \pm 0.14$   | $1.48 \pm 0.04$   | $2.75 \pm 1.73$   |
| Myopic   | $10000 \pm 2$     | $13437 \pm 9$     | $16935 \pm 35$    |
| DAD      | $1.22 \pm 0.09$   | $1.42 \pm 0.02$   | $1.93 \pm 0.20$   |
| RL-sBOED | $1.68 \pm 0.29$   | $3.31 \pm 1.33$   | $2.71 \pm 0.69$   |
| Step-DAD | $9113 \pm 7$      | $23243 \pm 24$    | $44693 \pm 32$    |

The offline timer includes initialization selection, optimization, periodic validation and checkpoint writes. DAD's reported stage time excludes the separately incurred Fixed

training used for initialization; a joint Fixed-DAD workflow must include both costs. Step-DAD reuses the matching DAD model: its offline entries repeat the DAD training cost, with no additional training run; refinement costs are online. Offline means and sample SDs use three training seeds. Random and Myopic require no offline training, so their entries are zero.

At  $T = 5$ , DAD requires 1.93 ms online per sequence, compared with 16.94 s for Myopic and 44.69 s for Step-DAD (Table VIII). Thus the near-identical DAD/Step-DAD scores have markedly different computational costs. Offline training amortizes design computation over repeated experiments, but this benefit depends on how often a trained policy is reused and on the physical deadline.

### C. Optimization and Estimator Diagnostics

Evaluation uses checkpoints selected by validation, rather than the final training update. A common update budget establishes neither equal convergence nor equal computational effort; DAD initialization selection also uses a separately trained Fixed sequence. Validation histories and source provenance are retained in the repository. Contrast-count and planner-particle sensitivity remain to be established, and a score below the sPCE ceiling does not itself demonstrate estimator accuracy.

### D. Scope and Limitations

The conclusions apply to the reduced lossless dynamics, independent parameter priors, and scalar sensor specified here. Training and evaluation share the same parameter distribution; robustness to prior shift, sensor bias, or unmodeled dynamics has not been established. Evaluation seed 1001 informed development, so the aggregate results are not wholly independent confirmation. Finite-contrast scoring and particle inference introduce errors distinct from ODE integration error. Finally, the zero-latency simulation does not establish online planners' real-time feasibility, and greater information gain does not by itself demonstrate safer control.

## VI. CONCLUSION

Sequential policy learning uses observation feedback to select informative power-system probes. Its full-sequence objective accounts for the value of later experiments, while offline training shifts design optimization away from execution.

On the studied reduced IEEE9 and IEEE14 models, DAD, RL-sBOED, and Step-DAD achieve higher mean information scores than Random, Fixed, and Myopic with three to five probes. Offline-trained DAD and RL-sBOED also greatly reduce online decision cost. Step-DAD offers modest additional gains on IEEE9 but no consistent improvement on IEEE14, with substantially greater online computation. These findings support adaptive probing for parameter identification; direct stability and control benefits remain to be tested. Future work should assess estimator sensitivity, model mismatch, and nonzero decision latency before drawing deployment conclusions.

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